Essentials Of A BATTERY MANAGEMENT SYSTEM

We take our batteries for granted and often misuse them, unknowingly. But there is much that goes behind designing a battery and, especially, its management system, without which it can run amuck. Read on to find out how the battery management systems are designed

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To a user a battery looks like a black-box, which magically produces electrical energy. And it looks like no active control is required for the battery to perform well. The truth, however, is that most batteries require some kind of electrical control to keep them from going bad and have a long working life. For some battery chemistries this control may be intermittent and simple, while for others it could be continuous and essential.

An intelligent system that takes care of the batteries is called battery management system (BMS). Here we shall discuss some essential parts of a BMS that apply to most types of batteries, and specifically to lithium-ion (Li-ion) based batteries.

A BMS manages the circulation of electrical energy to and from the battery and the distribution of this energy across its constituent cells. It tries to manage the charging and discharging of the battery to achieve some health goals for the battery. The most basic goals for a BMS are:

1. Always monitor the health of the battery
2. Control its charging and discharging to maximise battery health and performance
3. Protect the battery by maintaining its cells' voltage within the specifications
4. Report the battery's health status to the user

The BMS is a complete system and, like any other embedded system, it has following important parts:

1. Supervisory control element (microcontroller and/or application specific integrated circuit)